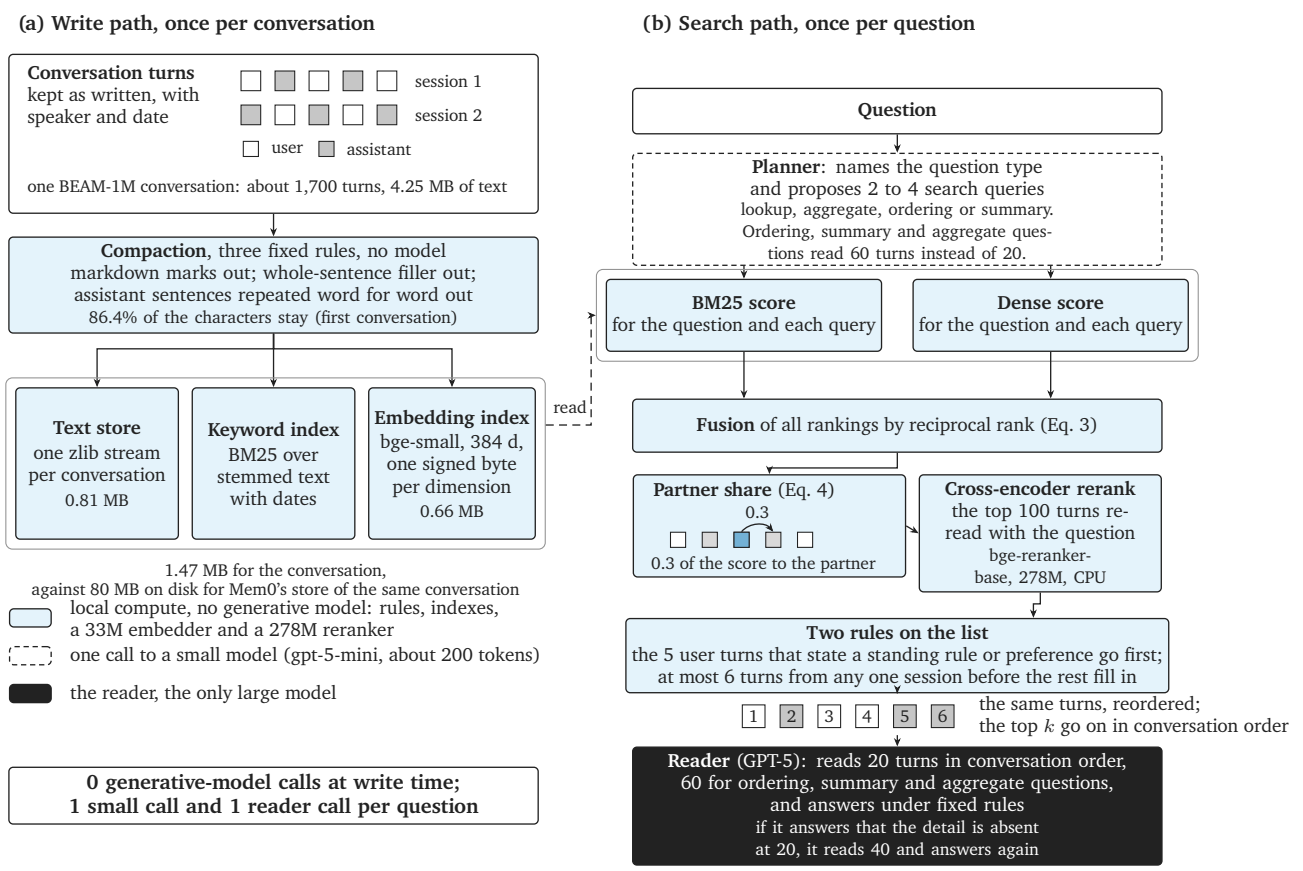
(a) Write path, once per conversation
Conversation turns
kept as written, with speaker and date
session 1
session 2
user
assistant
one BEAM-1M conversation: about 1,700 turns, 4.25 MB of text
Compaction, three fixed rules, no model
markdown marks out; whole-sentence filler out;
assistant sentences repeated word for word out
86.4% of the characters stay (first conversation)
Text store
one zlib stream per conversation
0.81 MB
Keyword index
BM25 over stemmed text with dates
Embedding index
bge-small, 384 d, one signed byte per dimension
0.66 MB
read
1.47 MB for the conversation,
against 80 MB on disk for Mem0's store of the same conversation
local compute, no generative model: rules, indexes, a 33M embedder and a 278M reranker
one call to a small model (gpt-5-mini, about 200 tokens)
the reader, the only large model
0 generative-model calls at write time;
1 small call and 1 reader call per question
(b) Search path, once per question
Question
Planner: names the question type and proposes 2 to 4 search queries
lookup, aggregate, ordering or summary.
Ordering, summary and aggregate questions read 60 turns instead of 20.
BM25 score
for the question and each query
Dense score
for the question and each query
Fusion of all rankings by reciprocal rank (Eq. 3)
Partner share (Eq. 4)
0.3
0.3 of the score to the partner
Cross-encoder rerank
the top 100 turns re-read with the question
bge-reranker-base, 278M, CPU
Two rules on the list
the 5 user turns that state a standing rule or preference go first;
at most 6 turns from any one session before the rest fill in
1 2 3 4 5 6
the same turns, reordered;
the top k go on in conversation order
Reader (GPT-5): reads 20 turns in conversation order,
60 for ordering, summary and aggregate questions,
and answers under fixed rules
if it answers that the detail is absent
at 20, it reads 40 and answers again